

Case 2: Structural Vector AutoRegressive Models, Local Projections and Macro-economic Policy Analysis

EBC4008 – Time Series Analytics

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Table of Contents

1. Data Exploration
2. Unit Root Testing
3. VAR Specification and Model Selection
4. Granger Causality
5. Impulse Response Functions

Data Exploration

Overview

- Data extracted from the Federal Reserve Economic Data (FRED)
- Selected series:
 - **INDPRO**: Industrial Production (Total Index)
 - **CPIAUCSL**: Consumer Price Index (All Urban Consumers)
 - **FEDFUNDS**: Effective Federal Funds Rate
- Original sample: January 1959 – December 2019
 - **Effective sample: March 1959 – December 2019** (...Why?)

Economic Interpretation

- The three series aim to capture distinct dimensions of aggregate economic activity
- Following *Bernanke (2005)*, they can be interpreted as:
 - *INDPRO* – Industrial Production → **Real activity / Output**
 - *CPIAUCSL* – Consumer Price Index → **Prices / Inflation**
 - *FEDFUNDS* – Effective Federal Funds Rate → **Monetary policy**

Benchmark Transformations

- *McCracken and Ng (2016)* provide benchmark transformations intended to obtain stationary representations of the FRED-MD series:

Series	Transformation	Interpretation
<i>INDPRO</i>	$\Delta \log(\text{INDPRO}_t)$	Output growth
<i>CPIAUCSL</i>	$\Delta^2 \log(\text{CPIAUCSL}_t)$	Change in inflation
<i>FEDFUNDS</i>	$\Delta \text{FEDFUNDS}_t$	Change in the policy rate

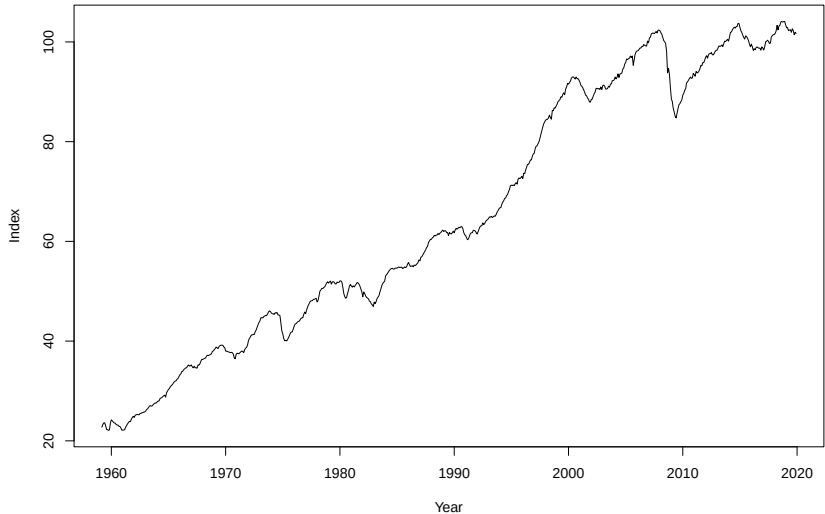
- These transformations are treated as benchmarks rather than assumptions

Series in Levels

- Before transforming the data, we inspect the original series over time

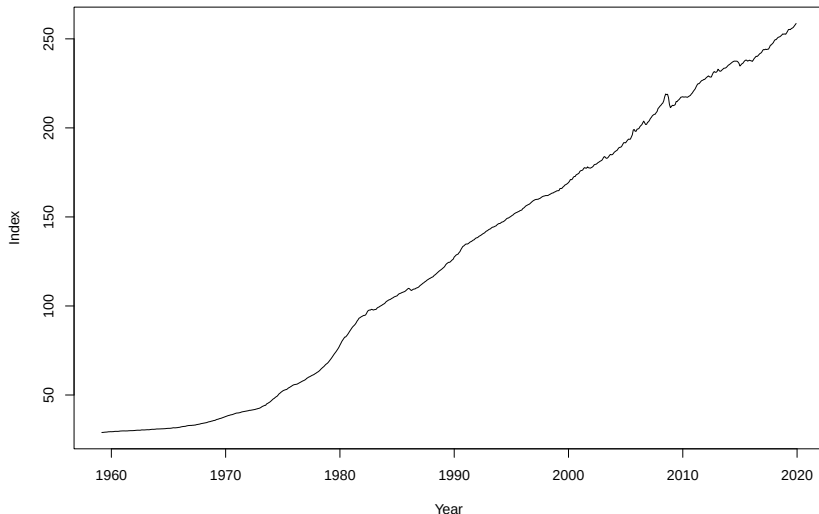
Series in Levels

Industrial Production



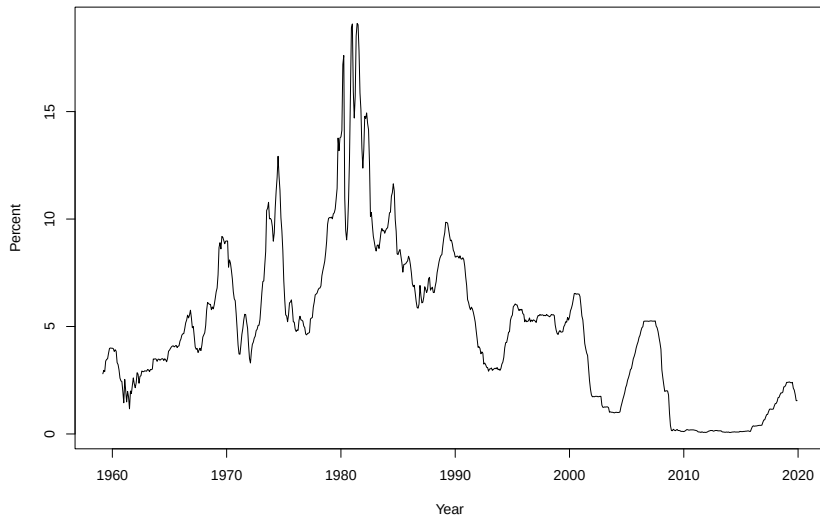
Series in Levels

Consumer Price Index



Series in Levels

Federal Funds Rate



Series in Levels

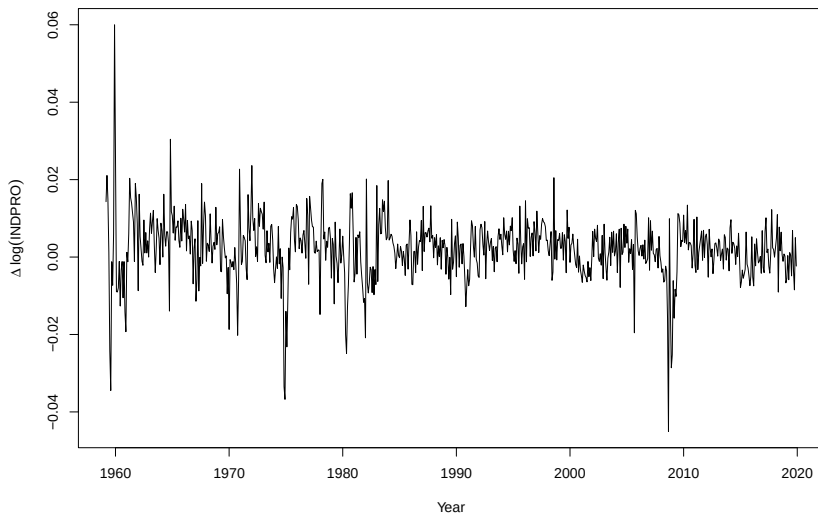
- Before transforming the data, we inspect the original series over time
 - *INDPRO* exhibits a pronounced long-run upward trend and cyclical fluctuations
 - *CPIAUCSL* displays a strong and persistent upward trend
 - *FEDFUNDS* shows substantial variation across the sample period
- The visual evidence suggests that stationarity cannot be assumed in levels

Transformed Series

- Now, we apply the FRED-MD benchmark transformations:
 - $\Delta \log(INDPRO_t)$
 - $\Delta^2 \log(CPIAUCSL_t)$
 - $\Delta FEDFUNDS_t$

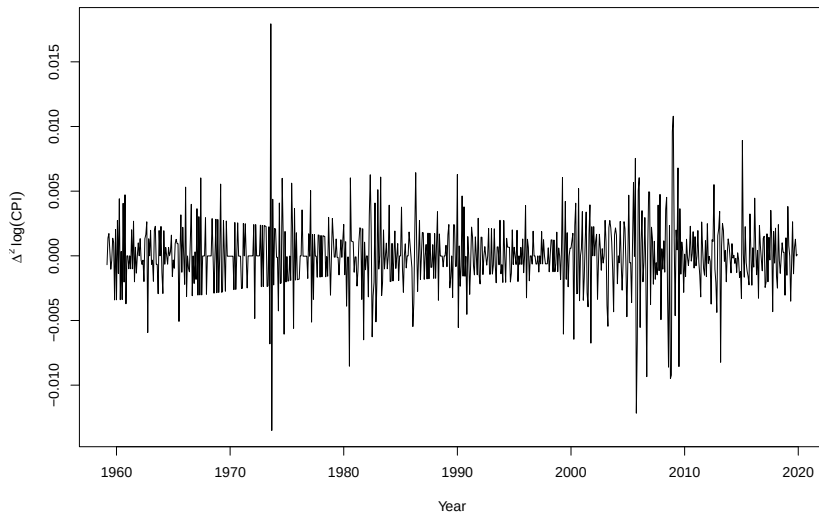
Transformed Series

Output Growth

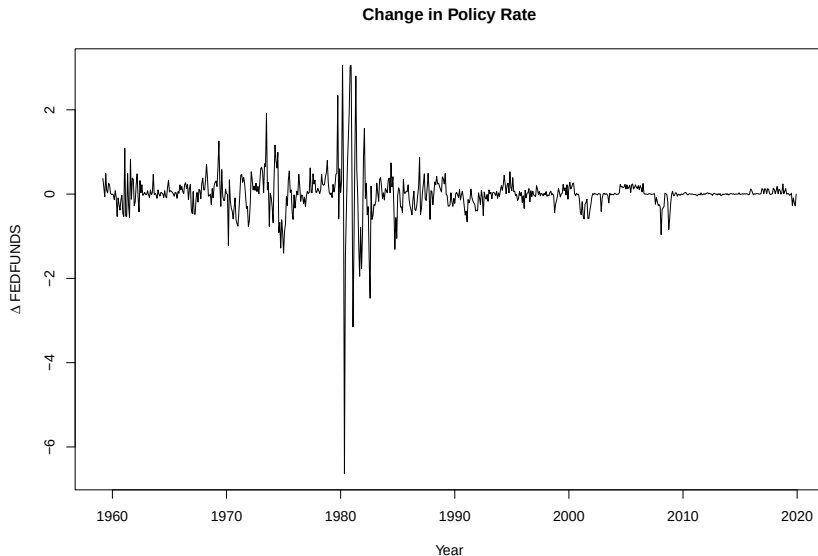


Transformed Series

Change in Inflation



Transformed Series



Transformed Series

- Now, we apply the FRED-MD benchmark transformations:
 - $\Delta \log(INDPRO_t)$
 - $\Delta^2 \log(CPIAUCSL_t)$
 - $\Delta FEDFUNDS_t$
 - The transformations substantially changes the persistence of the series
 - Less persistence and no obvious deterministic trend
 - Consistent with stationarity, but visual inspection is not sufficient to establish it
- ⇒ Still need to formally assess whether the transformed series are stationary

Unit Root Testing

Integrated processes

- The objective is to determine the order of integration of each series

$$y_t \sim I(d)$$

where d is the number of differences required to obtain a stationary series

- This determines the transformation used before estimating the VAR.

Manual Verification

- First, use `boot_adf()` directly to the level and successive differences

$$\Delta^2 y_t \rightarrow \Delta y_t \rightarrow y_t$$

Series	Level	Delta_1	Delta_2	Order
IP	0.828	<0.001	<0.001	1
CPI	0.838	0.048	<0.001	1
FFR	0.143	<0.001	<0.001	1

- $p < 0.05$ rejects the unit-root null; the integration order is the number of differences required for rejection

Automatic Integration Order

- We then apply `order_integration()` with bootstrap ADF tests

```
order_integration(  
  x, method = "boot_adf",  
  deterministics = "...", B = 4999)
```

- The procedure tests successive integration orders automatically

Series	Trend	Intercept
IP	1	1
CPI	2	1
FFR	1	1

- The procedure is repeated with a trend and with an intercept to assess sensitivity to the deterministic specification.

CPI: $I(1)$ or $I(2)$?

- CPI is sensitive to the deterministic specification
 - `order_integration()` with trend: $I(2)$
 - `order_integration()` with intercept: $I(1)$
 - Manual `boot_adf()` sequence: $I(1)$ at the 5% level
- The integration order of CPI is therefore not fully robust
- We retain the FRED-MD benchmark transformation $\Delta^2 \log(CPI)$ for the VAR

VAR Specification and Model Selection

VAR Specification

- Define the vector of stationary variables as

$$\mathbf{y}_t = \begin{bmatrix} IP_t \\ CPI_t \\ FFR_t \end{bmatrix} = \begin{bmatrix} \Delta \log(INDPRO_t) \\ \Delta^2 \log(CPIAUCSL_t) \\ \Delta FEDFUNDS_t \end{bmatrix}$$

- The VAR(p) can be written in companion form as

$$\xi_t = \mathbf{c}^* + \mathbf{F}\xi_{t-1} + \mathbf{v}_t$$

$$\xi_t = \begin{bmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix}, \quad \mathbf{v}_t = \begin{bmatrix} \mathbf{u}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_3 & 0 & \cdots & 0 & 0 \\ 0 & I_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_3 & 0 \end{bmatrix}$$

→ Here, ξ_t stacks the current and lagged values of the system

Lag Order Selection

- Common methods to determine the lag order given by *Kilian and Lütkepohl (2017)* include:
 - **Top-down testing:** Start with many lags and remove the highest lag step by step if it adds little information
 - ▶ *Adv.:* Simple and directly tests lag relevance
 - ▶ *Lim.:* Depends on the starting order
 - **Bottom-up testing:** Start with few lags and add more only if they meaningfully improve the model
 - ▶ *Adv.:* Favors parsimonious models
 - ▶ *Lim.:* May stop too early and miss longer-run dynamics
 - **Forecast-based selection:** Choose the lag order with the best predictive performance
 - ▶ *Adv.:* Directly targets forecasting
 - ▶ *Lim.:* Depends on horizon and evaluation sample

Information Criteria

- An alternative is to compare candidate VAR specifications using information criteria
 - **AIC** – Comparatively weaker penalty for additional parameters
 - **HQ** – Intermediate penalty for model complexity
 - **SC/BIC** – Stronger penalty, generally favoring more parsimonious models
 - **FPE** – Evaluates the estimated prediction error of competing specifications
- ▶ *Adv.:* Systematic comparison of fit and model complexity
- ▶ *Lim.:* Different criteria may select different lag orders

Lag Selection Results – Information Criteria

- Lag orders are selected using `VARselect()` from the `vars` package, comparing AIC, HQ, SC/BIC and FPE:

Criterion	Selected lag
AIC	14
HQ	4
SC / BIC	3
FPE	14

- AIC selects $p = 14$, while SC/BIC selects $p = 3$
⇒ These define the two candidate lag specifications

VAR Estimation

- The two candidate specifications are estimated using VAR()
 - AIC specification: VAR(14)
 - SC/BIC specification: VAR(3)
- For each lag j , estimation produces

$$\hat{A}_j = \begin{bmatrix} \hat{a}_{11,j} & \hat{a}_{12,j} & \hat{a}_{13,j} \\ \hat{a}_{21,j} & \hat{a}_{22,j} & \hat{a}_{23,j} \\ \hat{a}_{31,j} & \hat{a}_{32,j} & \hat{a}_{33,j} \end{bmatrix}$$

- Rows correspond to the *IP*, *CPI* and *FFR* equations; columns correspond to lagged *IP*, *CPI* and *FFR*
- Diagonal elements capture own-lag effects, while off-diagonal elements capture cross-variable predictive effects

VAR Stability

- Stability is assessed using `roots()` from the `vars` package
- A VAR is stable if all eigenvalues of the companion matrix satisfy $|\lambda_i| < 1$

Specification	Maximum root modulus
AIC	0.939
SC / BIC	0.665

- Both specifications satisfy the stability condition
 - AIC: maximum modulus = 0.939 – closer to the unit-circle boundary
 - SC/BIC: maximum modulus = 0.665 – farther inside the unit circle
- Hence, both estimated VARs are dynamically stable, although the AIC specification lies closer to the stability boundary

Residual Diagnostics

- Residual properties are assessed using `serial.test()`, `arch.test()` and `normality.test()` from the `vars` package (*Pfaff, 2008*)

- **Serial correlation:** H_0 – no residual serial correlation

- ▶ AIC VAR(14): $p = 0.0060 \rightarrow \text{reject } H_0$

- ▶ SC/BIC VAR(3): $p < 0.0001 \rightarrow \text{reject } H_0$

Residual dependence remains under both lag specifications

- **ARCH effects:** H_0 – no conditional heteroskedasticity

- ▶ Both specifications: $p < 0.0001 \rightarrow \text{reject } H_0$

Residual variance is not constant over time

- **(Normality:** H_0 – multivariate normality)

- ▶ Both specifications: $p < 0.0001 \rightarrow \text{reject } H_0$

Normality is not required for VAR estimation; its rejection motivates bootstrap-based inference for the subsequent IRFs

Model Selection

- Neither specification fully eliminates residual serial correlation
- The SC/BIC VAR(3) is more parsimonious, while the AIC VAR(14) allows for richer monthly dynamics
- VAR(14) is retained as the model specification for the subsequent IRF analysis
 - **Results should be interpreted cautiously given the remaining residual misspecification!**

Granger Causality

Granger Causality

- x Granger-causes y if past x improves the prediction of y , conditional on the VAR
- Granger causality is predictive, not necessarily structural causality
- The vars package provides the function `causality()`
 - For a some variable (or vector in this case) x ,

$$H_0 : \beta_{x_1,1} = \dots = \beta_{x_1,p} = \beta_{x_2,1} = \dots = \beta_{x_2,p} = 0$$

$$H_1 : \beta_{x_i,j} \neq 0 \quad \text{for at least one } j$$

Granger Causality

- In a trivariate VAR, `causality()` tests the chosen variables against the remaining one jointly

x	y	F_statistic	p_value	Decision
CPI + FFR	IP	1.50	0.044	Reject H0
IP + FFR	CPI	2.26	<0.001	Reject H0
IP + CPI	FFR	2.47	<0.001	Reject H0

Pairwise Granger Causality

- Test the six directional relationships separately while retaining the full trivariate VAR

x	y	F_statistic	p_value	Decision
IP	CPI	1.13	0.32647	Do not reject H0
IP	FFR	3.98	< 0.001	Reject H0
CPI	IP	1.48	0.11292	Do not reject H0
CPI	FFR	0.76	0.70957	Do not reject H0
FFR	IP	1.56	0.08426	Do not reject H0
FFR	CPI	2.15	0.00832	Reject H0

$$H_0 : \beta_{x,1} = \dots = \beta_{x,p} = 0$$

$$H_1 : \beta_{x,j} \neq 0 \quad \text{for at least one } j$$

Impulse Response Functions

Impulse Response Functions

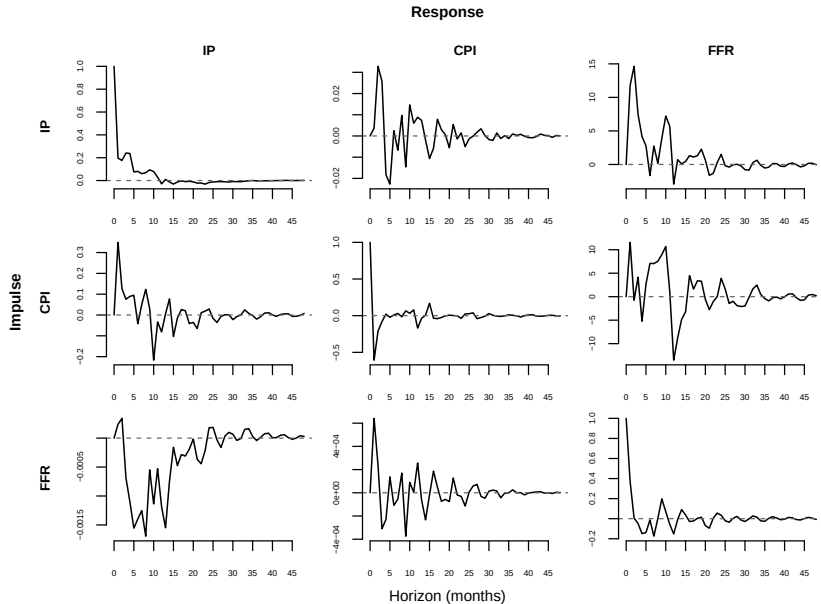
- An impulse response function traces how one variable reacts over time to an innovation in another variable
 - Impulse: variable receiving the initial innovation
 - Response: variable whose reaction is followed over subsequent horizons
 - The IRF summarizes the sign, magnitude and persistence of the response

IRF Implementation

- Compute IRFs using `irf()` from the `vars` package

```
irf(fit_aic,  
    impulse = variables,  
    response = variables,  
    n.ahead = 48,  
    ortho = FALSE,  
    boot = TRUE)
```

IRF Point Estimates



Responses to an IP Innovation

- Impulse: IP, $\Delta \log(INDPRO)$
 - IP — strongly positive on impact, followed by rapid decay and a small negative overshoot
 - CPI — alternating positive and negative movements that become progressively smaller
 - FFR — large positive short-run response followed by oscillations around zero
- Most of the effect is concentrated at short horizons and eventually dies out

Responses to a CPI Innovation

- Impulse: $CPI, \Delta^2 \log(CPI)$
 - IP — initially positive, followed by a negative overshoot and gradual convergence toward zero
 - CPI — positive on impact, followed by a sharp negative correction and mean reversion
 - FFR — large short-run fluctuations that diminish over time
- The responses are again concentrated mainly in the first 15–20 months

Responses to an FFR Innovation

- Impulse: FFR , ΔFFR
 - IP — becomes negative after a few months and gradually returns toward zero
 - CPI — oscillates around zero with declining amplitude
 - FFR — positive on impact, followed by a reversal and damped oscillations

The Price Puzzle

- Bernanke et al. (2005) discuss the price puzzle
 - A contractionary monetary-policy innovation (represented by an increase in the FFR) is followed by an increase in the price level
 - The relevant relationship in our VAR is an FFR impulse and a CPI response
- CPI enters the VAR as $\Delta^2 \log(CPI)$
 - The transformed CPI response therefore cannot by itself determine whether the price level rises

Bootstrap Confidence Intervals

- Point estimates do not show the sampling uncertainty surrounding the IRFs
- With `boot = TRUE`, `irf()` uses a residual bootstrap
 1. Estimate the VAR and obtain its residuals
 2. Resample centered residuals with replacement
 3. Generate a bootstrap time series recursively from the estimated VAR
 4. Re-estimate the VAR using the bootstrap sample
 5. Recompute the IRFs
 6. Repeat the procedure for all bootstrap replications

Constructing the Confidence Interval

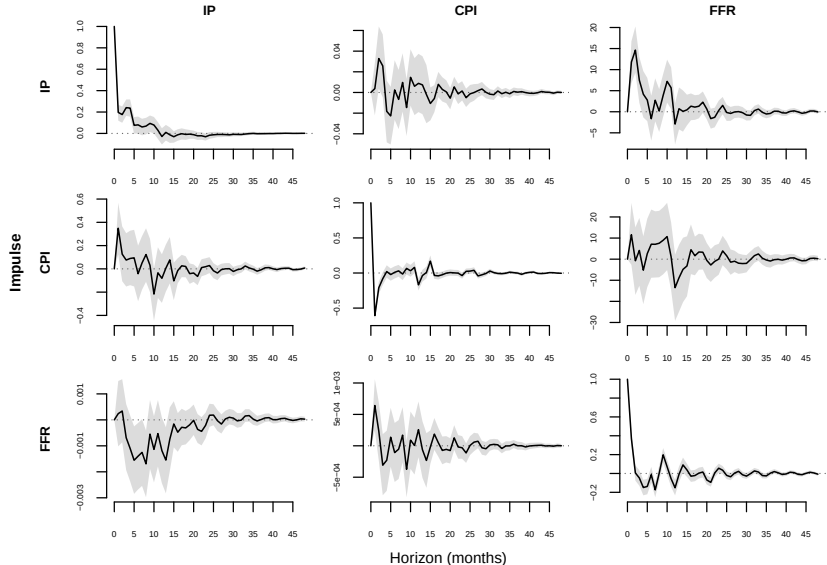
- Each bootstrap replication produces a new estimate of the IRF at every horizon
- The vars package uses a standard percentile interval

$$CI_h^{95\%} = [\hat{\theta}_{h,0.025}^*, \hat{\theta}_{h,0.975}^*]$$

- The lower and upper bounds are the 2.5th and 97.5th percentiles of the bootstrap IRF distribution at horizon h
- Pointwise confidence intervals: uncertainty is assessed separately at each horizon

IRFs with 95% Bootstrap Bands

Response



Significance of the IRFs

- A response is pointwise significant at the 5% level when its 95% confidence band excludes zero
 - The clearest evidence is concentrated in short-run own-responses
 - Several cross-responses are imprecisely estimated and their bands include zero over much of the horizon
 - Statistical significance generally weakens as the horizon increases

Recovering IRFs in Levels

- The VAR is estimated using stationary transformations rather than the original levels
- A response in first differences is converted to a level response by accumulating the IRF (Lütkepohl, 2005)
 - For industrial production:

$$IRF_h^{\log(INDPRO)} = \sum_{s=0}^h IRF_s^{\Delta \log(INDPRO)}$$

- For the federal funds rate:

$$IRF_h^{FFR} = \sum_{s=0}^h IRF_s^{\Delta FFR}$$

Recovering the CPI Level Response

- CPI enters the VAR in second log differences $\Delta^2 \log(CPI_t)$
- Its response therefore requires two accumulations

1. Recover the inflation response:

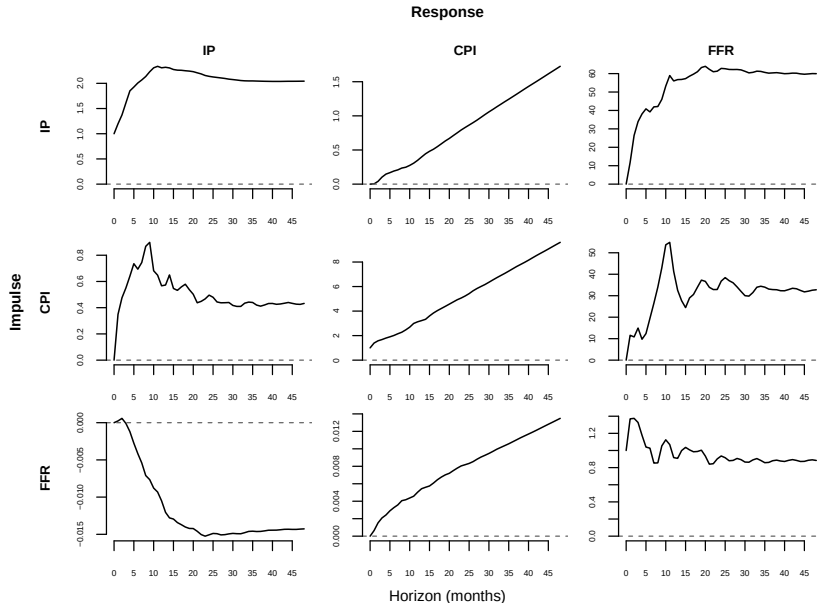
$$IRF_h^{\Delta \log(CPI)} = \sum_{s=0}^h IRF_s^{\Delta^2 \log(CPI)}$$

2. Recover the log price-level response:

$$IRF_h^{\log(CPI)} = \sum_{j=0}^h IRF_j^{\Delta \log(CPI)}$$

- The resulting CPI IRF describes the cumulative response of the log price level

IRFs Reconstructed in Levels



Interpretation of the Level Responses

- Accumulation changes the interpretation of the responses
 - An IP innovation is followed by higher industrial production and a higher CPI level over the horizon
 - A CPI innovation generates cumulative movements across all three variables in levels
 - An FFR innovation is associated with lower industrial production and a higher CPI level
- Unlike the IRFs of the differenced variables, the reconstructed level responses need not converge back to zero

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- ⇒ So, does the price puzzle occur?

Interpretation of the Level Responses

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 - An FFR innovation is associated with lower industrial production and a higher CPI level
 - Unlike the IRFs of the differenced variables, the reconstructed level responses need not converge back to zero
- ⇒ So, does the price puzzle occur? **Yes and no.**

Reduced Form and Structural Analysis

- Reduced-form IRFs describe responses to VAR forecast errors
- These innovations are not automatically interpretable as specific economic shocks
 - Reduced-form residuals may be contemporaneously correlated
 - A policy-rate innovation need not represent an exogenous monetary-policy shock
 - Structural analysis requires additional identifying restrictions

The present IRFs are informative about dynamic propagation, but causal structural interpretation requires an identified structural VAR

Thank you for your attention!